RONALD ANDREI S. PORTUGAL

Full-Stack Developer

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PROFESSIONAL SUMMARY

Full-stack developer with 3 years of experience building web applications for the government sector, including leading frontend work on an internal monitoring system. Works mainly with React, Next.js, TypeScript, Laravel and PostgreSQL, and is currently learning to build AI features with local LLMs and vector search.

WORK EXPERIENCE

Full-Stack Developer, AI-Powered Inventory System **Aug 2023 - Present**

Philippine Navy

Tech: Next.js, React, TypeScript, PostgreSQL, pgvector, Redis, Ollama, Vitest, GitHub Actions

- Built a Next.js and PostgreSQL inventory system for multiple offices, with stock tracking, item assignments, transfers, audit logs, CSV/PDF export and simple stock-out forecasts.
- Added an AI assistant that runs on a local Ollama model and answers inventory questions by combining keyword search with pgvector similarity search, showing links to the records it used.
- Tested the assistant with a set of sample questions; it found the right item in the top 3 results 97.5% of the time and answered 83% of questions correctly.
- Used PostgreSQL row-level security so each office only sees its own data, including through the AI assistant, and added email code sign-in, idle sign-out and rate limits.
- Fixed GitHub Actions type checks that failed on every run because Next.js route types were not generated on a clean machine.
- Wrote automated tests for office data separation and sign-in, and merged changes through a staging branch before main.

Full-Stack Developer, Awareness Reporting Platform

Philippine Navy

Tech: React, TypeScript, Node.js, PostgreSQL, Socket.IO, Docker, Caddy, nginx

- Developed a secure web platform for anonymous insider threat reporting, awareness surveys and administrative case review.
- Secured sensitive reports and evidence files with AES-256-GCM encryption, SHA-256 integrity checks and audit logging.
- Built password and email OTP sign-in with role-based access, account lockout and immediately revocable sessions.
- Created an admin dashboard with real-time notifications, charts, CSV export and account management.
- Improved performance by moving report encryption off the main request path and adding server-side search, filtering and paging.
- Reworked rate limits so personnel sharing one public IP could still submit reports and surveys while abuse protection stayed in place.
- Reduced security risk by adding a strict Content Security Policy, removing internal details from error messages and clearing known dependency vulnerabilities.
- Deployed the system with Docker, HTTPS, PostgreSQL and automated daily backups, and supported it with CI and automated tests.

Frontend Lead, Monitoring System

Philippine Navy

Tech: Laravel, Livewire, Flux UI, Tailwind CSS, PostgreSQL, Leaflet, ApexCharts, Ollama, pgvector

- Led frontend development of an internal system for encoding, reviewing, mapping and analyzing threat records across user roles.
- Coordinated with DevOps and backend developers to line up frontend changes with backend APIs and deployments.
- Redesigned the interface with shared page, table and chart components so every screen follows one layout, with loading skeletons and light and dark themes.

- Built an AI assistant on a self-hosted Ollama model with similarity search limited to records each user is authorized to see, citation checks and prompt logging.
- Reworked the map to fill the page with a floating toolbar and filters in one bar, so the controls cover far less of the map.
- Made dense map data readable by coloring points by category with a legend, per-user saved colors, clusters ringed by category share and detail modals.
- Created a dashboard widget showing each category's record count, pending reviews and status, linking directly to the filtered records and controlled by its own permission.
- Fixed cards and table rows lagging behind theme changes by suspending transitions during the switch, and fixed unreadable dropdown options in dark mode.
- Redesigned the sign-in screens and sign-in code email, blocked double submits, and added an interface size setting applied before first paint.
- Conducted manual testing of the system's screens and user flows to catch issues before changes were merged.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, PHP, SQL, HTML, CSS
Frontend: React, Next.js, Livewire, Flux UI, shadcn/ui, Tailwind CSS, Leaflet, ApexCharts
Backend: Node.js, Express, Laravel, Socket.IO, REST APIs
Databases: PostgreSQL, MySQL, Redis, Drizzle ORM
AI: Ollama, RAG, pgvector, LLM integration
Security: Role-based access control, OTP sign-in, encryption, rate limiting, audit logging
DevOps: Docker, GitHub Actions, CI/CD pipelines
Testing: Vitest, Jest, Pest, PHPStan, manual testing
Tools: Git, GitHub, GitLab, pgAdmin

EDUCATION

Bachelor of Science in Information Technology, Major in Web and Mobile Development **2019 – 2023**
Laguna State Polytechnic University, Los Baños Campus

REFERENCES

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